Blessed G/Michael Catholic Secondary School

Semester 1 Physics Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Feresula is a traditional unit to measure time.

____ **2.** The number 1.020 has four significant figures.

____ **3.** The number 1,000,000 is denoted by the metric prefix mega.

____ **4.** Zeros between non-zero digits are significant.

____ **5.** Leading zeros are significant if the number has a decimal point.

____ **6.** In addition, the rules for significant figures are the same as in division.

II. Choose the best answer among the given alternatives.

____ **7.** Which of the following is a fundamental physical quantity?

A. Area **B.** Volume **C.** Time **D.** Density

____ **8.** When adding or subtracting significant figures, the result should have the same number with:

A. the most decimal places **B.** the fewest significant figures

C. the fewest decimal places **D.** the most significant figures

____ **9.** The prefix "Nano" represents:

A. 10^{-6} **B.** 10^{6} **C.** 10^{9} **D.** 10^{-9}

____ **10.** Which one of the following innovations was contributed by Sir Isaac Newton?

A. universal law of gravitation **B.** Calculus

C. optics & laws of motion **D.** All of the above

____ **11.** Which one of the following numbers has the greatest number of significant figures?

A. 0.020 **B.** 2000 **C.** 0.001 **D.** 50000

____ **12.** The number 25.373 rounded to 3 significant figures is:

A. 25.30 **B.** 25 **C.** 25.3 **D.** 25.4

____ **13.** Multiply 500 × 2.4 and report with the correct number of significant figures:

A. 872 **B.** 8700 **C.** 2000.0 **D.** 1000

____ **14.** Subtract 35.25 - 34 and report with the correct number of significant figures:

A. 1.25 **B.** 1.5 **C.** 1 **D.** 2

____ **15.** Which one of the following is NOT equivalent to 1 hour?

A. 60 minutes **B.** 3600 seconds

C. 0.24 day **D.** None of the above

____ **16.** One metric ton is equivalent to which of the following?

A. 1000 kg **B.** 10 quintal

C. 1,000,000 g **D.** All of the above

____ **17.** Which physicist explained the photoelectric effect in 1905, establishing light quanta?

A. Albert Einstein **B.** Isaac Newton

C. James Prescott Joule **D.** Michael Faraday

____ **18.** Which scientist experimentally determined the mechanical equivalent of heat?

A. James Prescott Joule **B.** Marie Curie

C. Michael Faraday **D.** Albert Einstein

____ **19.** Which pioneering scientist discovered radium and polonium, winning two Nobel Prizes?

A. Marie Curie **B.** Isaac Newton

C. Michael Faraday **D.** James Prescott Joule

____ **20.** Which scientist discovered electromagnetic induction and formulated laws governing magnetic flux and induced EMF?

A. Michael Faraday **B.** Isaac Newton

C. Albert Einstein **D.** Marie Curie

____ **21.** Which branch of physics deals with the study of motion and forces?

A. Optics **B.** Electromagnetism

C. Mechanics **D.** Acoustics

____ **22.** Which branch of physics studies the behavior of heat, work, and temperature?

A. Thermodynamics **B.** Mechanics

C. Nuclear physics **D.** Astrophysics

____ **23.** Which branch of physics deals with electric and magnetic fields and their interactions?

A. Electromagnetism **B.** Acoustics

C. Optics **D.** Astrophysics

____ **24.** Which branch of physics is concerned with the generation, propagation, and detection of sound waves?

A. Acoustics **B.** Optics

C. Nuclear physics **D.** Mechanics

____ **25.** Which branch of physics studies the behavior, properties, and transmission of light?

A. Optics **B.** Thermodynamics **C.** Acoustics **D.** Mechanics

____ **26.** How many significant figures are in the measurement 30.250 m?

A. 2 **B.** 3 **C.** 4 **D.** 5